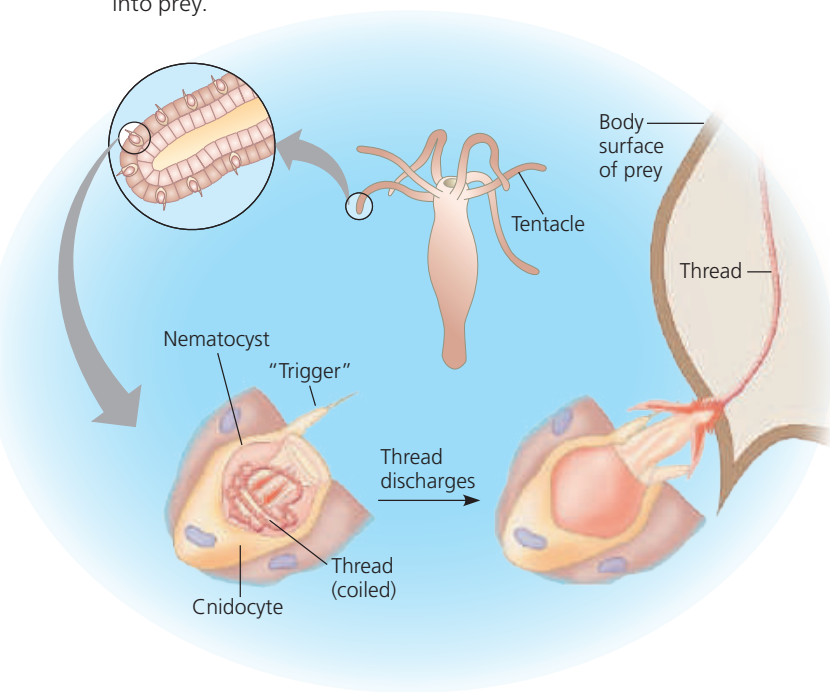


▼ **Figure 33.5 A cnidocyte of a hydra.** This type of cnidocyte contains a stinging capsule, the nematocyst, which contains a coiled thread. When a “trigger” is stimulated by touch or by certain chemicals, the thread shoots out, puncturing and injecting poison into prey.



gastrodermis (inner layer) have bundles of microfilaments arranged into contractile fibers. The gastrovascular cavity acts as a hydrostatic skeleton (see Concept 50.6) against which the contractile cells can work. When a cnidarian closes its mouth, the volume of the cavity is fixed, and contraction of selected cells causes the animal to change shape. Cnidarians have no brain. Instead, movements are coordinated by a noncentralized *nerve net* that is associated with sensory structures distributed around the body. Thus, even though it lacks a brain, the animal can detect and respond to stimuli from all directions.

Fossil and molecular evidence suggests that early in its evolutionary history, the phylum Cnidaria diverged into two major clades, Medusozoa and Anthozoa (**Figure 33.6**).

Medusozoans

All cnidarians that produce a medusa are members of clade Medusozoa, a group that includes the *scyphozoans* (jellies) and *cubozoans* (box jellies) shown in Figure 33.6a, along with the *hydrozoans*. Most hydrozoans alternate between the polyp and medusa forms, as seen in the life cycle of *Obelia* (**Figure 33.7**). The polyp stage, a colony of interconnected polyps in the case of *Obelia*, is more conspicuous than the medusa. Hydras, among the few cnidarians found in fresh water, are also unusual hydrozoans in that they exist only in polyp form.

Unlike hydrozoans, most scyphozoans and cubozoans spend the majority of their life cycles in the medusa stage. Coastal scyphozoans, for example, often have a brief polyp stage during their life cycle, whereas those that live in the open ocean generally lack the polyp stage altogether. As their name (which means “cube animals”) suggests, cubozoans have a box-shaped medusa stage. Most cubozoans live in tropical oceans and are equipped with highly toxic cnidocytes. For example, the sea wasp (*Chironex fleckeri*), a cubozoan that lives off the coast of northern Australia, is one of the deadliest organisms known: Its sting causes intense pain and can lead to respiratory failure, cardiac arrest, and death within minutes.

▼ **Figure 33.6 Cnidarians.**

(a) Medusozoans



Many scyphozoans (jellies) are bioluminescent. Food captured by nematocyst-bearing tentacles is transferred to specialized oral arms for transport to the mouth.



This cubozoan, a sea wasp, produces a poison that can subdue fish, crustaceans (as seen here), and other large prey. The poison is more potent than cobra venom.

(b) Anthozoans



Sea anemones and other anthozoans exist only as polyps. Many anthozoans form symbiotic relationships with photosynthetic algae.



These star corals live as colonies of polyps. Their soft bodies are enclosed at the base by a hard exoskeleton.

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